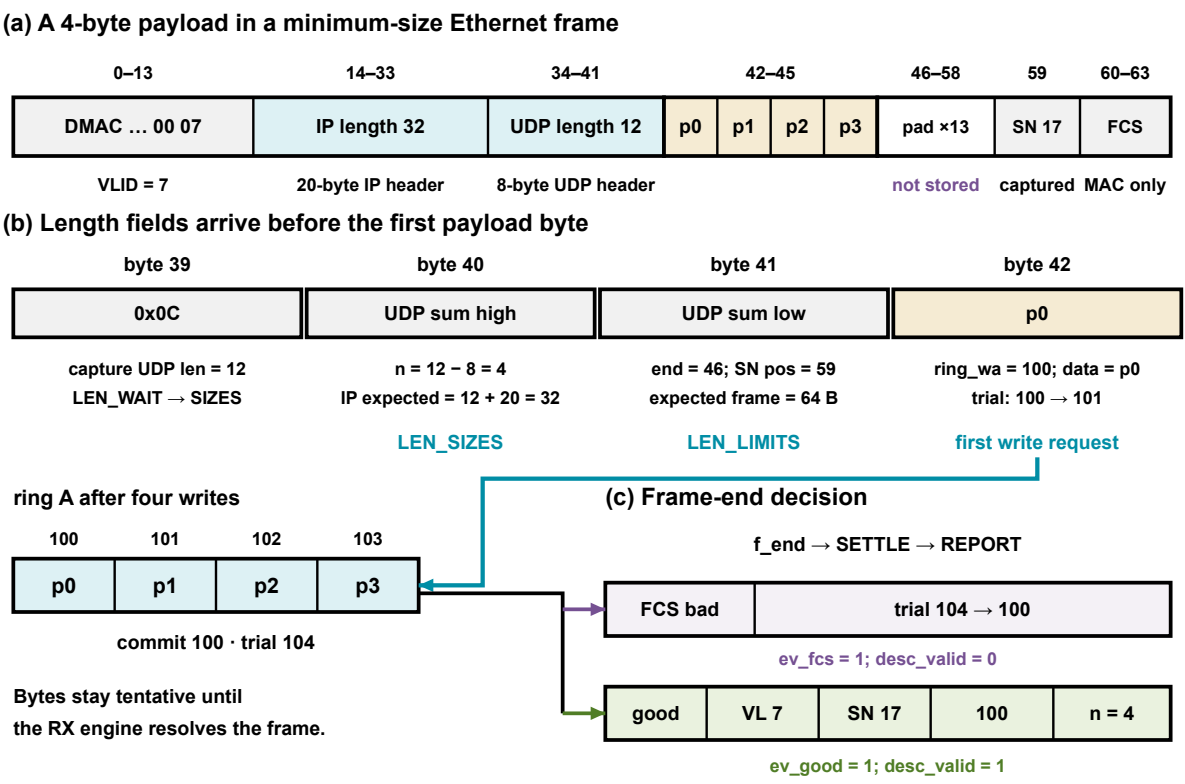
(a) A 4-byte payload in a minimum-size Ethernet frame
0–13
14–33
34–41
42–45
46–58
59
60–63
DMAC … 00 07
IP length 32
UDP length 12
p0
p1
p2
p3
pad ×13
SN 17
FCS
VLID = 7
20-byte IP header
8-byte UDP header
not stored
captured
MAC only
(b) Length fields arrive before the first payload byte
byte 39
byte 40
byte 41
byte 42
0x0C
UDP sum high
UDP sum low
p0
capture UDP len = 12
LEN_WAIT → SIZES
n = 12 − 8 = 4
IP expected = 12 + 20 = 32
end = 46; SN pos = 59
expected frame = 64 B
ring_wa = 100; data = p0
trial: 100 → 101
LEN_SIZES
LEN_LIMITS
first write request
ring A after four writes
100
101
102
103
p0
p1
p2
p3
commit 100 · trial 104
Bytes stay tentative until
the RX engine resolves the frame.
(c) Frame-end decision
f_end → SETTLE → REPORT
FCS bad
trial 104 → 100
ev_fcs = 1; desc_valid = 0
good
VL 7
SN 17
100
n = 4
ev_good = 1; desc_valid = 1